

MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

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Abstract—We propose MinImageScorer, an error-driven framework that scores per-object detection difficulty from baseline model failures, combining confidence and localization penalties calibrated via Pearson correlation optimization. Validated on MinneApple and Global Wheat Head (GWHD), the score reliably ranks objects and images by difficulty (Pearson r up to 0.82 with miss rate) and recovers domain- and size-level failure patterns without supervision. Score-guided copy-paste outperforms random selection on MinneApple ($\Delta\text{AP}=+0.013$) but both strategies fall below baseline due to distributional shift. On GWHD, domain shift renders copy-paste ineffective under either strategy. These results show that MinImageScorer correctly identifies hard objects, and that copy-paste utility is determined by the dataset’s failure mode, not the selection strategy.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift

I. INTRODUCTION

Agricultural object detection benchmarks such as MinneApple [1] and Global Wheat Head (GWHD) [2] exhibit distinct structural failure modes: MinneApple is dominated by sub-resolution small objects, while GWHD has severe cross-domain visual variation. Copy-paste augmentation has demonstrated gains on MS COCO [3], [4], where neither condition is present. Applying it blindly to structurally different datasets risks corrupting the training distribution rather than improving it, yet no prior work provides an empirical framework to diagnose this mismatch.

We address this gap with three contributions: (1) **MIS** (MinImageScorer), a per-object difficulty score derived from model failure signals with data-driven weight calibration; (2) empirical validation showing the score correctly recovers image-, domain-, and object-size-level difficulty without supervision; and (3) a controlled three-seed study showing score-guided copy-paste outperforms random selection on a resolution-limited dataset, while both strategies fail on a domain-shifted dataset.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ be ground-truth objects in image I and $\mathcal{P}(g) = \{p : \text{IoU}(p, g) \geq \tau\}$ the matched predictions. Denoting $c_p = \text{conf}(p)$ and $u_p = \text{IoU}(p, g)$, the *object difficulty score* is:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset, \\ \alpha + \beta, & \text{otherwise,} \end{cases} \quad (1)$$

where α and β weight confidence and localization penalties. Missed objects receive the maximum score $\alpha + \beta$. The *image-level score* combines mean object difficulty with false-positive rate:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPRate}(I). \quad (2)$$

Object scoring uses low-confidence predictions ($\tau=0.01$) to capture near-misses; FP rate uses optimal-threshold predictions to reflect deployment errors. Weights α, β, w_1, w_2 are calibrated by maximizing Pearson correlation with per-image miss rate and FP rate on the training set, requiring no domain-specific assumptions.

B. Score-Guided Copy-Paste Pipeline

A baseline detector is trained with all built-in augmentations disabled. Inference on the training set yields confidence and IoU signals used to compute S_{obj} and S_{img} . For score-guided augmentation, objects are sampled proportionally to S_{obj} via exponential weighting; for random, sampling is uniform. Both conditions paste 2–8 objects per image and retrain on the augmented set under identical hyperparameters.

III. SCORE VALIDATION

A. Correlation with Detection Failures

Table I reports Pearson correlation between S_{img} and per-image miss rate and FP rate on training images. All cases

TABLE I
PEARSON CORRELATION OF S_{img} WITH PER-IMAGE MISS RATE AND FP RATE (TRAINING SET, W/ TRADITIONAL AUGMENTATION).

Dataset	$r(\text{score, miss rate})$	$r(\text{score, FP rate})$	Inter-seed r
MinneApple	0.698	0.688	0.806
GWHD	0.816	0.818	0.900

show strong positive correlation ($r \geq 0.69$). GWHD with traditional augmentation reaches $r=0.82$ on both metrics, where the baseline model is stable and the score signal is clean. On MinneApple without traditional augmentation, inter-seed stability drops to $r \approx 0.67$: without color jitter and flipping, unstabilized models produce fluctuating confidence on densely clustered small objects, making the score training-run dependent. This is a property of baseline instability, not a flaw in the scoring design.

B. Domain- and Size-Level Validity

Beyond per-image correlation, the score recovers structural difficulty without access to test labels. On GWHD, mean S_{img} inversely tracks AP across all seven source domains (Spearman $\rho = -0.93$): the hardest domain (*arvalis 2*, AP=0.365) scores 0.431, while the easiest (*rres 1*, AP=0.572) scores only 0.260. On MinneApple, small objects (≤ 32 px) average a score of 0.378 vs. 0.222 for medium objects, exactly matching the AP gap (0.132 vs. 0.522). These results confirm that MIS captures image-, domain-, and object-size-level difficulty from training-set model failures alone.

IV. AUGMENTATION EXPERIMENTS

A. Setup

All models use YOLOv8s with COCO pretrained weights. Training: MinneApple 200 epochs / patience 10; GWHD 100 epochs / patience 8; batch 16; input 1024×1024 ; lr 10^{-2} . All built-in augmentations are disabled to isolate copy-paste. Augmentation ratio 0.3; scale jitter [0.9, 1.1] (MA) and [0.8, 1.2] (GWHD). All results are 3-seed means.

B. Results

Table II shows that on MinneApple, score-guided copy-paste outperforms random across all sub-metrics ($\Delta\text{AP} = +0.013$), confirming the score directs augmentation toward higher-signal objects. However, both conditions fall below baseline (-0.024 for Score CP). The AP₇₅ drop (0.452 $\rightarrow$ 0.440) indicates degraded localization precision from pasting hard objects without blending or masks; the AP_S drop (0.300 $\rightarrow$ 0.286) indicates score-selected objects are predominantly sub-resolution and cannot be made more learnable by repetition.

On GWHD, score-guided and random selection are indistinguishable ($\Delta\text{AP} = -0.002$) and both fall below baseline (-0.023). Score-guided sampling concentrates objects from domain-minority sources (*arvalis*), amplifying rather than correcting the existing domain imbalance. Backbone feature analysis (YOLOv8s P4, UMAP) confirms that augmented images are displaced from the test distribution relative to

REPLACE with scatter: S_{img} vs. miss rate
Left: MinneApple ($r=0.70$) Right: GWHD ($r=0.82$)
Each point = one training image. Regression line shown.

Fig. 1. S_{img} vs. per-image miss rate. Higher score predicts harder images on both datasets. Points generated from training-set inference only.

TABLE II
THREE-SEED MEAN COCO AP. **BOLD** = BEST COPY-PASTE CONDITION PER DATASET.

Dataset	Condition	AP	AP ₅₀	AP ₇₅	AP _S	AP _M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	0.428	0.759	0.440	0.286	0.581
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	0.484	0.905	0.455	0.098	0.477

original training images on both datasets, with score-guided images showing greater displacement on GWHD— directly explaining the consistent degradation.

V. CONCLUSION

MIS reliably quantifies detection difficulty from model failure signals, validated by per-image correlations (r up to 0.82), near-perfect domain difficulty ranking (Spearman $\rho = -0.93$ on GWHD), and correct object-size ordering on MinneApple—all without test labels. Score-guided copy-paste outperforms random selection on MinneApple (+0.013 AP) but both strategies degrade below baseline due to distributional shift between augmented and test images. On GWHD, domain shift makes copy-paste ineffective regardless of selection quality. The practical implication is direct: copy-paste is effective for object underrepresentation (COCO-like settings) but not for resolution-limited or domain-shifted datasets; MIS provides the diagnostic to determine which condition holds before augmentation is applied. Future work includes domain-constrained pasting and online augmentation to reduce distributional rigidity.

REFERENCES

- [1] N. Hani, P. Roy, and B. Bhanu, “MinneApple: A Benchmark Dataset for Apple Detection and Segmentation,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 852–858, 2020.
- [2] E. David *et al.*, “Global Wheat Head Detection (GWHD) Dataset,” *Plant Phenomics*, vol. 2021, 2021.
- [3] M. Kisantal *et al.*, “Augmentation for Small Object Detection,” in *Proc. ICACCI*, 2019.
- [4] G. Ghiasi *et al.*, “Simple Copy-Paste is a Strong Data Augmentation Method for Instance Segmentation,” in *Proc. IEEE/CVF CVPR*, pp. 2918–2928, 2021.